

A Jai-inspired systems language in Rust. Incremental salsa front end, lossless CST, typed SSA mid-end, two execution engines held byte-identical by a differential harness.

6/6 gates green 919 tests ADR-0064 latest W3 in progress · temporary storage + backtraces left

| TESTS                  | CORPUS                 | ADRS                    | DIAGNOSTICS            | EDITOR CHECKS              |
|------------------------|------------------------|-------------------------|------------------------|----------------------------|
| 919                    | 148                    | 64                      | 88                     | 156                        |
| workspace, all passing | jr files, both engines | 0001 to 0064, immutable | codes, E0258 next free | Neovim, verified not gated |

The vertical slice, and what it costs to be honest about it

|                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>EXIT CRITERIA</p> <p><b>DONE</b> hello.jr runs in the VM</p> <p><b>DONE</b> compiles to a native binary</p> <p><b>DONE</b> rustc-grade diagnostics</p> <p><b>DONE</b> formatter round-trips the corpus</p> <p><b>DONE</b> editor integration (Neovim)</p> <p><b>OPEN</b> verified Linux x86-64 CI run</p> <p>The last one needs a push. Configured, never run, so it is a decision rather than a technical gap.</p> | <p>HOW CORRECTNESS IS ESTABLISHED</p> <p><b>Both engines run every corpus program</b> and must agree, on output and on trap wording, from one shared formatter.</p> <p><b>MIR snapshots pin the exact instruction sequence.</b> This wave the pointer-arithmetic file shows <code>p + n</code> as a slot holding the pointer, indexed — the same load-then-scale a view’s data word takes, so both back ends scale by the element stride and no size appears in the mid-end.</p> <p><b>The differential harness catches regressions the wave did not aim at.</b> Typing pointer operands with no expectation broke <code>sum: s64 = xx tiny + 1;</code> — the autocast lost its <code>s64</code> context. A guard that skips the pointer probe when a numeric type is expected fixed it; the harness found it because a passing corpus file suddenly failed to compile.</p> <p><b>How a feature lowers can narrow its scope.</b> <code>p - q</code>, the pointer difference, read as obviously part of “pointer arithmetic” until the lowering showed its element-count result needs the stride — layout the mid-end deliberately does not carry. Cut mid-wave and deferred, rather than smuggled in with a layout query <code>jr-mir</code> lacks.</p> |
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The language today

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| <p>WORKS, END TO END</p> <p>Full integer tower, bool, string, pointers</p> <p>float32 and float64, plain IEEE-754, no traps</p> <p>struct, nominal, one level</p> <p>union, nominal, untagged, fields at offset 0</p> <p>enum and enum_flags, namespaced, bare dot-member</p> <p>Fixed arrays and views, bounds-checked, returnable</p> <p>cast and xx from context; operator overloading</p> <p>Trapping arithmetic, wrapping variants, bitwise</p> <p>if, else, while, for, break, continue, defer, using</p> <p>Multiple returns, named args, literal defaults</p> <p>import, foreign, system_library, #scope_module</p> <p>One trivial compile-time run, folded</p> <p>Traps name their source line</p> <p>Bounds checks, strippable by build setting or #no_abc</p> <p>context, a hidden parameter passed by pointer</p> <p>Procedures as values: call, pass, return, struct field</p> <p>null, a context-typed pointer literal; malloc and free</p> <p>An allocator in the context, installed and called through</p> <p>Pointer offset <code>p + n</code>, <code>n + p</code>, <code>p - n</code> — element-scaled, unchecked</p> <p>push_context, a block with its own copy of the context</p> | <p>ABSENT (WAVE)</p> <p>pointer difference <code>p - q</code>, deferred</p> <p>percent on floats, <code>is_nan</code>, math (W7)</p> <p>a tagged variant type (W4.5)</p> <p>explicit backing type; a switch (W4.5)</p> <p>array literals, sub-slicing</p> <p>unary, index and call overloading</p> <p>transmute; float printing</p> <p>#must; a multi-result call in a return</p> <p>polymorphs, macros (W5)</p> <p>arbitrary run, RTTI, insert (W4)</p> <p>backtraces (W3)</p> <p>a per-index #no_abc; any other build setting</p> <p>temp storage (W3)</p> <p>a cross-file or foreign proc value; comparing one</p> <p>pointer arithmetic; cast to a pointer</p> <p>temp storage (W3)</p> <p>the difference <code>p - q</code>; <code>p[n]</code>; ordering</p> |
| <p>No error-handling model yet — ADR-0008 reserves the slot and the first half exists (several return values); #must is owed its own ADR. No GC and no RAII, which is a design value rather than a gap. A union is untagged, so reading a field other than the one last written reinterprets bits: documented in three places because it cannot be diagnosed. No indirect calls, which is the single largest gap in the project and is what blocks the rest of W3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Compiler internals, 17 crates

| STAGE                             | STATE       | NOTE                                                             |
|-----------------------------------|-------------|------------------------------------------------------------------|
| Lexer, parser, CST, typed AST     | works       | Hand-written, error-recovering, trivia-preserving                |
| Formatter                         | works       | Pure function over the CST; lost source in 7 of the last 8 waves |
| HIR, name resolution, modules     | works       | Flat import merge; cycles legal; export filtering                |
| InternPool: types, values, layout | works       | One layout computation and one integer evaluator, shared         |
| Sema: signatures, checking        | works       | E0212 to E0257; no const-eval here, by design                    |
| MIR: typed SSA                    | works       | Block parameters, notphis; explicit bounds check and zeroing     |
| Mid-end                           | 5 passes    | Inline, forwarding, const-prop, DCE, plus the bounds-check strip |
| Const-eval                        | works       | Runs MIR through the bytecode VM                                 |
| VM: register bytecode, libffi     | works       | No JIT; indirect calls, and malloc from its own region           |
| Cranelift back end                | works       | Aggregate returns via sret; indirect calls via func_addr         |
| LLVM back end                     | not started | W8 owns it                                                       |
| Language server                   | 12 caps     | Diagnostics, hover, goto, completion, rename, actions, hints     |
| Neovim integration                | works       | Runtimepath dir, no plugin manager; 156 checks                   |
| Driver                            | stub        | Should consume the workspace notion that now exists              |

Roadmap: W1 and W2 closed, W3 started

|                        |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W1 Data                | done        | Numeric tower, wrapping ops, enum, enum_flags, union, arrays, views, cast, xx, operator overloading. Closed by ADR-0048.                                                                                                                                                                                                                                                                                                                                                                 |
| W2 Flow and scope      | done        | for with it and it_index, labelled break, defer, using, multiple returns, named and default arguments, #scope_module. Closed by ADR-0054; ADR-0055 then closed a gap six corpus files had carried for eleven waves.                                                                                                                                                                                                                                                                      |
| W3 Runtime core        | in progress | Seven of eight done. context (ADR-0057), the bounds-check build setting (ADR-0058, finishing ADR-0003), indirect calls (ADR-0059), null plus a memory source (ADR-0060/0061), the allocator protocol (ADR-0062), push_context (ADR-0063), and pointer arithmetic (ADR-0064) — p + n on a *T, element-scaled and unchecked, so a bump allocator can advance a pointer. Remaining: temporary storage (now unblocked — the bump allocator it needed is writable) and traps with backtraces. |
| W4 Comptime            | not started | Arbitrary compile-time execution, RTTI and Type values, insert, code. PLAN section 5 names this the project's top risk: sema and comptime become mutually recursive.                                                                                                                                                                                                                                                                                                                     |
| W4.5 Pattern matching  | not started | switch with exhaustiveness, a bare dot-member as a case, and a tagged variant type. Was missing from the wave table entirely — two accepted ADRs deferred decisions to it while no wave scheduled it.                                                                                                                                                                                                                                                                                    |
| W5 Polymorphism        | not started | Polymorphic parameters, modify, bake_arguments, expand macros with hygiene, instantiation caching.                                                                                                                                                                                                                                                                                                                                                                                       |
| W6 Metaprogram         | not started | Workspaces, the compiler message loop, build scripts replacing makefiles, note attributes.                                                                                                                                                                                                                                                                                                                                                                                               |
| W7 Stdlib              | not started | Written in Jairs: String, dynamic array, hash table, Sort, Math, Random, File.                                                                                                                                                                                                                                                                                                                                                                                                           |
| W8 Performance         | not started | LLVM via inkwell for release builds, inliner maturity, struct-of-arrays, SIMD, parallel sema. Also owns the compile-throughput number, still unpublished.                                                                                                                                                                                                                                                                                                                                |
| W9 Tooling depth       | not started | Semantic tokens are the one LSP capability still missing; the other twelve landed early.                                                                                                                                                                                                                                                                                                                                                                                                 |
| W10 Graphics, in Jairs | not started | Window creation through foreign calls, Metal then Vulkan, immediate-mode 2D.                                                                                                                                                                                                                                                                                                                                                                                                             |

What this session did, and how each wave was scoped

**SIXTEEN WAVES SHIPPED**

ADR-0049 through 0064: for and defer, using, aggregate returns, multiple returns, named and default arguments, scope visibility, imported constants, float constants, context, the bounds-check build setting, indirect calls, null plus a memory source, the allocator protocol, push\_context, and pointer arithmetic.

Test count 900 to 919. Corpus 116 to 148 files. Neovim checks 103 to 156. W2 closed; W3 opened and seven of its eight features landed. A program can install an allocator in the context, scope it with push\_context, and advance a raw pointer — so a bump allocator is finally writable.

**Each wave was scoped by writing the feature and seeing what the compiler refused**, not by reading the handoff. That found gaps the handoff had missed in both directions: a proc-pointer struct field it called absent (it worked), and a void-returning proc-pointer type it never mentioned (unspellable, and blocking).

**TWO CLAIMS THE CODE DISPROVED**

**An ADR corrected mid-wave.** ADR-0060 §4 asserted the VM dereferences a malloc'd pointer via libffi. Running the corpus file the same ADR promised would pass disproved it: the VM's memory is a linear region and a host address is not an offset into it. ADR-0061 corrects it — the VM allocates from its own region, native calls libc, the bits differ and nothing observes them.

**A diagnostic that could not be acted on.** Installing an imported #foreign procedure into an allocator field reported “expected (s64) -> \*u8, found (s64) -> \*u8” — the same text twice, because the types differ only in an invisible calling convention. It is E0256 now, the code that says to wrap it.

**THE THING WORTH YOUR DECISION**

Fourteen waves sit uncommitted across five branches. Two waves ago a careless git checkout reverted the tree-sitter grammar nine waves and cost an hour's reconstruction — the concrete price of not committing. Committing each wave as it goes green would bound the damage from any future slip to a single wave.

*Recorded as a recommendation, not a change: the authorisation is yours to give.*

Sources: PLAN.md §1.5 and §7, the ADR directory, docs/decisions/DECISIONS.md, and the six gates run today. Every number was measured rather than carried forward — the test count from a full workspace run, the corpus count from a file walk, the diagnostic codes from the const E0nnn definitions across the crates, and the editor-check count from a real verify.lua run. Rebuild with typst compile jairs-dashboard.typ.